Advanced Statistics: Sampling and Frequency Distribution Examples

Part 1: Sampling Examples

Question 1: Random Sampling

A university has 2,000 students. The statistics department wants to estimate the average number of hours students spend on homework weekly. A simple random sample of 50 students is selected.

Step-by-step Solution:

1. Define the population: $X = \{x_1, x_2, \ldots, x_{2000}\}$, where x_i is the weekly homework hours of student i.

2. Select a random sample $S \subset X$ of size $n = 50$, where each student has equal probability $p = \frac{50}{2000} = 0.025$ of being chosen.

3. Calculate the sample mean:

$$\bar{X}_{sample} = \frac{1}{50} \sum_{x_i \in S} x_i$$

4. Estimate the population mean:

$$\bar{X}_{population} \approx \bar{X}_{sample}$$

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Question 2: Cluster Sampling

A city has 120 schools. The education board wants to estimate the average math test score for 8th graders. 10 schools are selected randomly, and all 8th graders in these schools are surveyed.

Solution:

1. Divide the population into clusters: $C_1, C_2, \ldots, C_{120}$, where C_j contains all students in school j.

2. Randomly select 10 clusters: $S_C \subset \{C_1, \ldots, C_{120}\}$.

3. Include all students in the selected clusters in the sample:

$$S = \bigcup_{C \in S_C} C$$

4. Compute the sample mean:

$$\bar{X}_{sample} = \frac{1}{|S|} \sum_{x_i \in S} x_i$$

5. Estimate population mean:

$$\bar{X}_{population} \approx \bar{X}_{sample}$$

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Question 3: Stratified Sampling

A company has 500 employees: 200 in admin, 150 in technical, and 150 in marketing. They want to estimate average monthly work hours. A stratified sample of 10% from each department is taken.

Solution:

1. Divide population into strata: $X = X_{\text{admin}} \cup X_{\text{tech}} \cup X_{\text{marketing}}$
2. Sample 10% from each stratum:

$$n_{\text{admin}} = 20, \quad n_{\text{tech}} = 15, \quad n_{\text{marketing}} = 15$$

3. Compute sample means for each stratum:

$$\bar{X}_{\text{admin}}, \quad \bar{X}_{\text{tech}}, \quad \bar{X}_{\text{marketing}}$$

4. Compute weighted population estimate:

$$\bar{X}_{population} = \frac{200}{500}\bar{X}_{\text{admin}} + \frac{150}{500}\bar{X}_{\text{tech}} + \frac{150}{500}\bar{X}_{\text{marketing}}$$

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Question 4: Systematic Sampling

A factory produces 10,000 items per month. To inspect quality, every 100th item is selected starting from a random item between 1 and 100.

Solution:

1. Define the population: $X = \{x_1, x_2, \dots, x_{10000}\}$
2. Select step size: $k = 100$
3. Choose a random starting point $r \in [1, 100]$
4. Sample:

$$S = \{x_r, x_{r+100}, x_{r+200}, \dots\}$$

5. Compute defect rate from the sample:

$$\hat{p} = \frac{\text{defective items in } S}{|S|}$$

Part 2: Frequency Distribution and Graphs

Example 1: Continuous Variable (Histogram & Ogive)

A class of 60 students has test scores ranging 35–100. Construct a frequency table with 10-point intervals.

Solution:

Interval	Frequency	Cumulative Frequency	Relative Cumulative Frequency
30–39	2	2	0.033
40–49	4	6	0.100
50–59	8	14	0.233
60–69	12	26	0.433
70–79	16	42	0.700
80–89	10	52	0.867
90–100	8	60	1.000

Step-by-step:

1. Count number of students in each score interval for frequency.
2. Compute cumulative frequency by summing frequencies up to each interval.
3. Compute relative cumulative frequency by dividing cumulative frequency by total number of students.
4. Histogram: Plot intervals (x-axis) vs. frequency (y-axis).
5. Ogive: Plot cumulative frequency against upper interval boundaries.

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Example 2: Discrete Variable (Bar Chart & Pie Chart)

Number of books read by 50 students in a month:

Books	Frequency
0	5
1	8
2	12
3	10
4	9
5	6

Solution:

1. Count the number of students for each number of books read.
2. Bar chart: Books on x-axis, frequency on y-axis.
3. Pie chart: Calculate proportion of each category: Proportion $= \frac{\text{Frequency}}{50}$

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Example 3: Bivariate Frequency (Line Chart & Scatter Plot)

Weights (kg) and heights (cm) of 10 students:

Height	150	155	160	165	170	175	180	185	190	195
Weight	45	50	52	55	60	63	70	75	80	85

Solution:

1. Create intervals for weight (e.g., 45–49, 50–54, etc.) and count frequencies if needed.
2. Line chart: Plot weight vs. height to observe trends.
3. Scatter plot: Plot individual (height, weight) points to visualize correlation.